

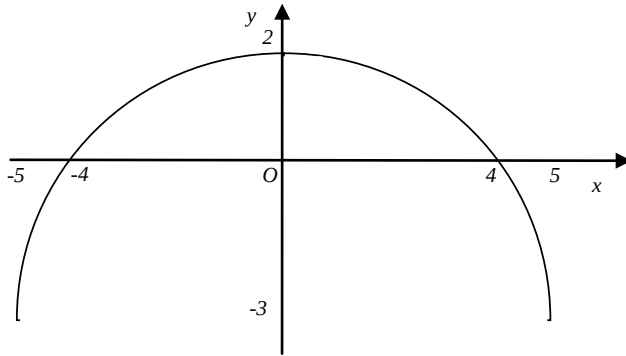
QUESTION 1: START A NEW PAGE

	Marks
(a) Evaluate: $\int_0^{\frac{\pi}{2}} \cos^7 x \sin x dx$	2
(b) Find $\int \frac{dx}{(x+1)(x^2+2)}$	4
(c) Use the substitution $t = \tan \frac{\theta}{2}$ to evaluate $\int_0^{\frac{\pi}{3}} \frac{1}{1 + \sin \theta} d\theta$	4
(d) (i) Find the equations to the two tangents to the hyperbola $\frac{x^2}{4} - y^2 = 1$ which are parallel to the line $2x - y = 3$.	3
(ii) Show that the distance between these two tangents is $2\sqrt{3}$ units.	2

QUESTION 2: START A NEW PAGE

Marks

- (a) The diagram shows the graph of $y = f(x)$ for $-5 \leq x \leq 5$.



Draw separate sketches of the graphs of the following for $-5 \leq x \leq 5$.

(i) $y = \frac{1}{f(x)}$ 2

(ii) $y = f(x) - |f(x)|$ 2

(iii) $y = e^{f(x)}$ 2

(iv) $y = [f(x)]^2$ 2

- (b) Points $P(a \cos \theta, b \sin \theta)$ and $Q(a \cos \phi, b \sin \phi)$ lie on the ellipse

$$\frac{x^2}{4} + \frac{y^2}{3} = 1, \quad \text{where } \phi = \theta + \frac{\pi}{2}.$$

(i) Show that the gradient of the tangent at Q is $\frac{\sqrt{3} \sin \theta}{2 \cos \theta}$. 2

(ii) If α is the acute angle between the tangents at P and Q prove that 3

$$\tan \alpha = \frac{4\sqrt{3}}{|\sin 2\theta|}$$

(iii) Find the value of $OP^2 + OQ^2$ where O is the centre of the ellipse. 2

QUESTION 3: START A NEW PAGE

- (a)** (i) If $I_n = \int \sin^n x dx$ show that: **Marks**
3

$$I_n = -\frac{1}{n} \cos x \sin^{n-1} x + \left(\frac{n-1}{n} \right) I_{n-2} \text{ for } n \geq 2$$

- (ii) Hence find $\int \sin^5 x dx$. **3**

- (b)** Points $P(a \cos \theta, b \sin \theta)$, $Q(a \cos(\theta + \phi), b \sin(\theta + \phi))$ and $R(a \cos(\theta - \phi), b \sin(\theta - \phi))$ are three points on the ellipse **4**

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Show that OP bisects the chord QR , where O is the origin.

- (c)** (i) Use the substitution of $u = \pi - x$ to show: **2**

$$\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{x}{\sin x} dx = \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{\pi - x}{\sin x} dx$$

- (ii) Evaluate $\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{x}{\sin x} dx$ **3**

[You may assume $\int \operatorname{cosec} \theta d\theta = \ln |\operatorname{cosec} \theta - \cot \theta|$]

QUESTION 4: START A NEW PAGE

- | | Marks |
|---|--------------|
| (a) Evaluate: $\int_0^2 x e^{2x} dx$ | 2 |
| (b) (i) Sketch the following graphs for $-2\pi \leq x \leq 2\pi$ on the same number plane without using calculus. | 2 |
| (α) $y = \frac{x}{2}$ | |
| (β) $y = \sin x$ | |
| (ii) Sketch on a separate number plane the graph of | 2 |
| $y = \frac{2 \sin x}{x}$ for $-2\pi \leq x \leq 2\pi$. | |
| (c) A hyperbola has the equation: | |
| $\frac{x^2}{3} - \frac{y^2}{2} = 1$. | |
| (i) Find the coordinates of the foci. | 2 |
| The equation of the tangent to the hyperbola at $P(\sqrt{3} \sec \theta, \sqrt{2} \tan \theta)$ is: | |
| $\sqrt{2}x \sec \theta - \sqrt{3}y \tan \theta = \sqrt{6}$. | |
| (Do not prove this equation) | |
| (ii) Show that the equation of the normal at P is: | 2 |
| $\sqrt{3}x \tan \theta + \sqrt{2}y \sec \theta = 5 \sec \theta \tan \theta$. | |
| The tangent and normal to the hyperbola at P cut the y axis at T and N respectively. | |
| (iii) Show that the circle with TN as diameter passes through the foci of the hyperbola. | 5 |

END OF EXAMINATION

(a)

$$\int_0^{\pi/2} \cos^3 x \sin x \, dx$$

$$= - \int_0^{\pi/2} -\sin x \cos^3 x \, dx$$

This is of the form: $-\int f'(x)[f(x)]^n \, dx$

$$= \left[-\frac{\cos^3 x}{3} \right]_0^{\pi/2}$$

$$= -0 - \left(-\frac{1}{3}\right)$$

$$= \frac{1}{3}$$

① For correct integration

① For correct substitution and answer

(b)

Write as partial fractions

$$\frac{dx}{(x+1)(x^2+2)} = \left[\frac{A}{x+1} + \frac{Bx+C}{x^2+2} \right] dx$$

$$\therefore 1 = A(x^2+2) + (Bx+C)(x+1)$$

$$= Ax^2 + 2A + Bx^2 + Bx + Cx + C$$

Equate coefficients

$$A+B=0 \quad \therefore A=-B$$

$$B+C=0 \quad \therefore C=-B$$

$$2A+C=1 \quad \therefore -2B-B=1$$

$$\therefore B = -\frac{1}{3}$$

$$A = \frac{1}{3}$$

$$C = \frac{1}{3}$$

① Writing as partial fractions and producing simultaneous equations

① for correct solution of equations

$$\frac{1}{3} \int \left(\frac{1}{x+1} + \frac{-x+1}{x^2+2} \right) dx$$

$$= \frac{1}{3} \int \left(\frac{1}{x+1} - \frac{\frac{1}{2}(2x)}{x^2+2} + \frac{1}{x^2+2} \right) dx$$

$$= \frac{1}{3} \left[\ln|x+1| - \frac{1}{2} \ln|x^2+2| + \tan^{-1} \frac{x}{\sqrt{2}} \right]$$

$$= \frac{1}{3} \left[\ln \left(\frac{x+1}{\sqrt{x^2+2}} \right) + \tan^{-1} \frac{x}{\sqrt{2}} \right] + C$$

① for integration of terms producing log functions

① for integration to $\tan^{-1} \frac{x}{\sqrt{2}}$

(c)

$$\int_0^{\pi/3} \frac{1}{1+\sin \theta} \, d\theta$$

$$t = \tan \frac{1}{2} \theta$$

$$\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{1}{2} \theta$$

$$= \frac{1}{2} [1+t^2]$$

$$\theta = \frac{\pi}{3} \quad t = \frac{1}{\sqrt{3}}$$

$$\theta = 0 \quad t = 0$$

$$\int_0^{\frac{1}{\sqrt{3}}} \frac{1}{1+\left(\frac{2t}{1+t^2}\right)} \left(\frac{2}{1+t^2}\right) dt$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2}{1+t^2+2t} dt$$

$$= \int_0^{\frac{1}{\sqrt{3}}} \frac{2}{(1+t)^2} dt$$

$$= \left[\frac{-2}{(1+t)} \right]_0^{\frac{1}{\sqrt{3}}}$$

$$= -2 \left[\left(\frac{1}{1+\frac{1}{\sqrt{3}}} \right) - \frac{1}{1} \right]$$

$$= -2 \left[\frac{\sqrt{3}}{\sqrt{3}+1} - \frac{(\sqrt{3}+1)}{(\sqrt{3}+1)} \right]$$

$$= \frac{-2}{(\sqrt{3}+1)} (-1)$$

$$= \frac{2}{\sqrt{3}+1}$$

$$= \frac{2(\sqrt{3}-1)}{(\sqrt{3}+1)(\sqrt{3}-1)}$$

$$= \frac{2}{\sqrt{3}-1}$$

① correct derivative in $\frac{dt}{d\theta}$ and change of limit values

① correct change of variable in integral

① correct integration

① correct substitution and simplified answer

(rationalised denominator not necessary for full marks)

1 (d) (i) $\frac{x^2}{4} - y^2 = 1$

tangent parallel to $2x - y = 3$
must be in the form
 $y = 2x + c$

$\therefore \frac{x^2}{4} - (2x + c)^2 = 1$

$\frac{x^2}{4} - (4x^2 + 4xc + c^2) = 1$

$\therefore x^2 - 16x^2 - 16xc - 4c^2 = 4$

$\therefore 15x^2 + 16xc + 4(c^2 + 1) = 0$

as line is a tangent $\therefore \Delta = 0$

$\therefore \Delta = (16c)^2 - 4 \times 15 \times 4(c^2 + 1)$

$0 = 16^2 c^2 - 16 \times 15(c^2 + 1)$

$0 = 16c^2 - 15c^2 - 15$

$c^2 = 15$

$c = \pm \sqrt{15}$

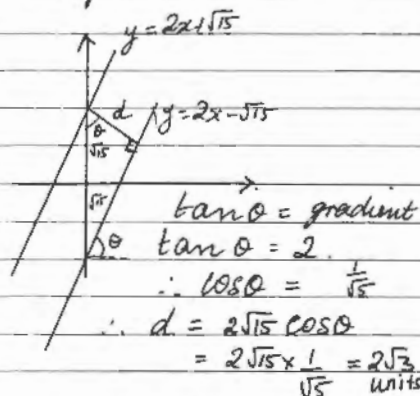
$\therefore$ tangents are: $y = 2x \pm \sqrt{15}$

Alternative solutions using

$$xx_1 - yy_1 = 0$$

are also possible.

(ii)



Alternative solutions using
perp distance formula

① correct substitution
of $y = 2x + c$

① correct Δ

① correct tangent
equations

① Derivative/gradient

① Equation of tangent
to hyperbola

① Equation of tangents

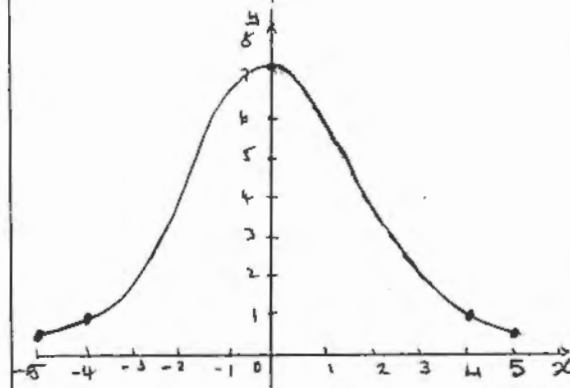
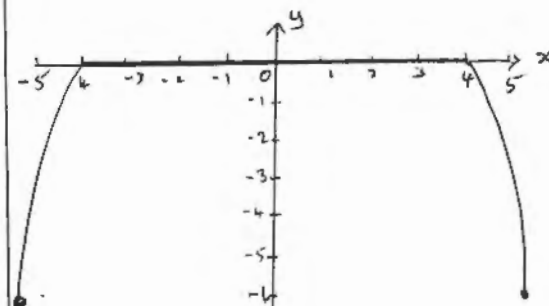
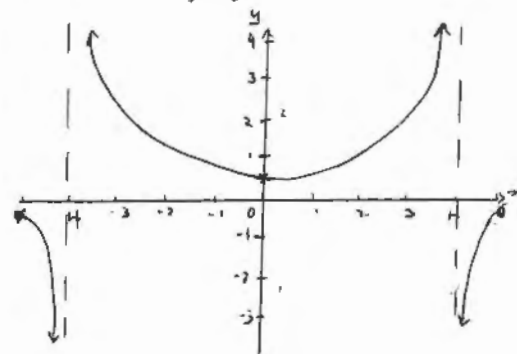
① calculation of
 $\cos \theta$

① correct distance

① point on line

① correct distance

2(a) (i) $y = \frac{1}{f(x)}$



① asymptotes +
intercepts

① shape

① $y = 0$ $-4 \leq x < 4$

① graph for
 $-5 \leq x \leq 4$ and
 $+4 \leq x \leq 5$

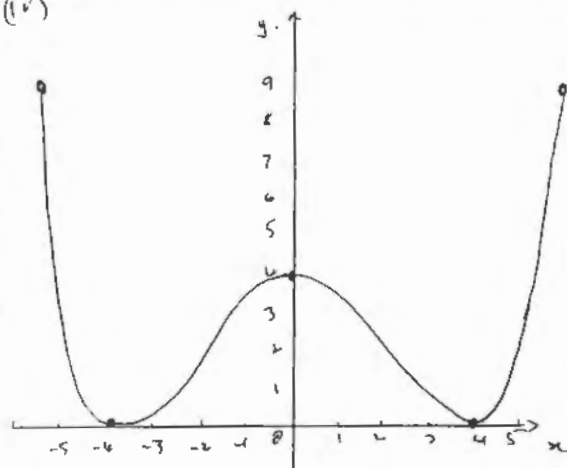
① values for $x = 0$
and $x = \pm 1$

① shape

5

2(a)

(1v)

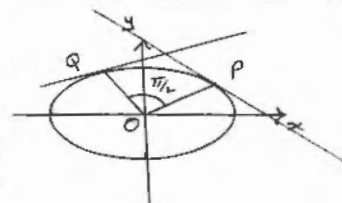


$$y = [f(x)]^2$$

- ① Endpoints and intercepts
- ② Shape.

6

2(b)



(i)

$$\frac{x^2}{4} + \frac{y^2}{3} = 1 \quad a=2, \quad b=\sqrt{3}$$

$$\frac{2x}{4} + \frac{2y}{3} \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{2x}{4} \times \frac{3}{2y} = -\frac{3x}{4y}$$

grad of tangent at Q

$$\begin{aligned} m &= -\frac{3}{4} \frac{a \cos \phi}{b \sin \phi} \\ &= -\frac{3}{4} \frac{2 \cos(\theta + \pi/2)}{\sqrt{3} \sin(\theta + \pi/2)} \\ &= -\frac{\sqrt{3}}{2} \times \frac{-\sin \theta}{\cos \theta} \\ &= \frac{\sqrt{3}}{2} \frac{\sin \theta}{\cos \theta} \end{aligned}$$

(ii)

gradient of tangent at P is

$$\begin{aligned} \tan \alpha &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{\frac{\sqrt{3}}{2} \frac{\sin \alpha}{\cos \alpha} - \frac{-\sqrt{3} \cos \alpha}{\sin \alpha}}{1 + \frac{\sqrt{3}}{2} \frac{\sin \alpha}{\cos \alpha} \times \frac{-\sqrt{3} \cos \alpha}{\sin \alpha}} \right| \\ &= \left| \frac{-\frac{\sqrt{3}}{2} [\sin^2 \alpha + \cos^2 \alpha]}{\cos \alpha \sin \alpha (1 - \frac{3}{4})} \right| \\ &= \frac{\frac{\sqrt{3}}{2} (1)}{\frac{1}{2} \sin 2\alpha [\frac{1}{4}]} = \frac{4\sqrt{3}}{|\sin 2\alpha|} \end{aligned}$$

① derivative

① correct substitution and proof.

① correct gradient of tangent at P

① correct substitution into formula.

① correct simplification

(7)

$$\begin{aligned}
 OP^2 &= a^2 \cos^2 \theta + b^2 \sin^2 \theta \\
 OQ^2 &= a^2 \cos^2 \phi + b^2 \sin^2 \phi \\
 &= a^2 \cos^2 (\theta + \pi/2) + b^2 \sin^2 (\theta + \pi/2) \\
 &= a^2 \sin^2 \theta + b^2 \cos^2 \theta \\
 \therefore OP^2 + OQ^2 &= a^2 (\cos^2 \theta + \sin^2 \theta) + b^2 (\sin^2 \theta + \cos^2 \theta) \\
 &= a^2 + b^2 \\
 &= 4 + 3 \\
 &= 7
 \end{aligned}$$

① OP^2 and OQ^2 equations

① calculations

4(a) (i) $I_n = \int \sin^n x dx$

$$\begin{aligned}
 &= -\cos x \sin^{n-1} x + \int \cos x (n-1) \sin^{n-2} x \cos x dx \\
 &= -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x \cos^2 x dx \\
 &= -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x (1 - \sin^2 x) dx \\
 &= -\cos x \sin^{n-1} x + (n-1) \int \sin^{n-2} x dx - (n-1) \int \sin^n x dx \\
 &= -\cos x \sin^{n-1} x + (n-1) I_{n-2} - (n-1) I_n
 \end{aligned}$$

① conversion to I_{n-1} and I_{n-2}

$$\begin{aligned}
 I_n (1 - (n-1)) &= -\cos x \sin^{n-1} x + (n-1) I_{n-2} \\
 n I_n &= -\cos x \sin^{n-1} x + (n-1) I_{n-2} \\
 I_n &= -\frac{1}{n} \cos x \sin^{n-1} x + \frac{(n-1)}{n} I_{n-2}
 \end{aligned}$$

① correct proof of answer

(ii) $\int \sin^5 x dx = I_5$

$$\begin{aligned}
 I_5 &= -\frac{1}{5} \cos x \sin^4 x + \frac{4}{5} I_3 \\
 I_3 &= -\frac{1}{3} \cos x \sin^2 x + \frac{2}{3} I_1 \\
 I_1 &= \int \sin x dx \\
 &= -\cos x + C
 \end{aligned}$$

① correct sub into I_n for $n=5, 3, 1$

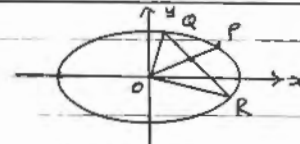
$$\begin{aligned}
 I_5 &= -\frac{1}{5} \cos x \sin^4 x + \frac{4}{5} \left[-\frac{1}{3} \cos x \sin^2 x \right] + \frac{4}{5} \times \frac{2}{3} [-\cos x] + C \\
 &= -\frac{1}{5} \cos x \sin^4 x - \frac{4}{15} \cos x \sin^2 x - \frac{8}{15} \cos x + C
 \end{aligned}$$

① correct addition of terms.

① Answer

(8)

3(b)



$P(a \cos \theta, b \sin \theta)$

$Q(a \cos (\theta + \phi), b \sin (\theta + \phi))$

$R(a \cos (\theta - \phi), b \sin (\theta - \phi))$

Midpoint QR = $M(x_1, y_1)$

$$x_1 = \frac{a}{2} (\cos (\theta + \phi) + \cos (\theta - \phi))$$

$$= \frac{a}{2} [\cos \theta \cos \phi - \sin \theta \sin \phi + \cos \theta \cos \phi + \sin \theta \sin \phi]$$

$$= \frac{a}{2} [2 \cos \theta \cos \phi]$$

$$= a \cos \theta \cos \phi$$

$$y_1 = \frac{b}{2} [\sin (\theta + \phi) + \sin (\theta - \phi)]$$

$$= \frac{b}{2} [\sin \theta \cos \phi + \cos \theta \sin \phi + \sin \theta \cos \phi - \cos \theta \sin \phi]$$

$$= \frac{b}{2} [2 \sin \theta \cos \phi]$$

$$= b \sin \theta \cos \phi$$

① x co-ord midpoint QR

① y co-ord midpoint QR

Gradient OP = $\frac{b \sin \theta}{a \cos \theta}$

Equation of OP $y = \frac{b \sin \theta}{a \cos \theta} x$

Sub $M(x_1, y_1)$ $x_1 = a \cos \theta \cos \phi$

$$RHS = \frac{b \sin \theta}{a \cos \theta} \times a \cos \theta \cos \phi$$

$$= b \sin \theta \cos \phi$$

$$= y_1$$

$\therefore$ OP passes through midpoint QR

$\therefore$ OP bisects $\angle QOR$

① equation of OP

① correct test showing OP passes through midpoint QR

(9)

3(c)

$$\int_{\pi/3}^{2\pi/3} \frac{x}{\sin x} dx$$

$$u = \pi - x$$

$$\frac{du}{dx} = -1$$

$$x = \frac{2\pi}{3} \quad u = \frac{\pi}{3}$$

$$x = \frac{\pi}{3} \quad u = \frac{2\pi}{3}$$

$$\therefore \int_{\pi/3}^{2\pi/3} \frac{x}{\sin x} dx = \int_{2\pi/3}^{\pi/3} \frac{(\pi-u)(-1)du}{\sin(\pi-u)}$$

$$= \int_{2\pi/3}^{\pi/3} \frac{\pi-u}{\sin u} du$$

$$\text{(change of variable)} \quad u = x \quad = \int_{\pi/3}^{2\pi/3} \frac{\pi-x}{\sin x} dx$$

$$\therefore \int_{\pi/3}^{2\pi/3} \frac{x}{\sin x} dx = \int_{\pi/3}^{2\pi/3} \frac{\pi-x}{\sin x} dx$$

$$\int_{\pi/3}^{2\pi/3} \frac{x}{\sin x} dx = \int_{\pi/3}^{2\pi/3} \frac{\pi}{\sin x} dx - \int_{\pi/3}^{2\pi/3} \frac{x}{\sin x} dx$$

$$\therefore 2 \int_{\pi/3}^{2\pi/3} \frac{x}{\sin x} dx = \pi \int_{\pi/3}^{2\pi/3} \frac{1}{\sin x} dx$$

$$= \pi \int_{\pi/3}^{2\pi/3} \operatorname{cosec} x dx$$

$$\int_{\pi/3}^{2\pi/3} \frac{x}{\sin x} dx = \frac{\pi}{2} \left[\ln |\operatorname{cosec} x - \cot x| \right]_{\pi/3}^{2\pi/3}$$

$$= \frac{\pi}{2} \left[\ln \left| \frac{\operatorname{cosec} \frac{2\pi}{3} - \cot \frac{2\pi}{3}}{\operatorname{cosec} \frac{\pi}{3} - \cot \frac{\pi}{3}} \right| \right]$$

$$= \frac{\pi}{2} \ln \left[\frac{\frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}}{\frac{2}{\sqrt{3}} - \frac{1}{\sqrt{3}}} \right]$$

$$= \frac{\pi}{2} \ln 3$$

① derivative and change of variable
limit values

① correct substitution and change of variable

① splitting + combining integrals

① integration and substitution of values

① simplified answers

(10)

4(a)

$$\int_0^2 x e^{2x} dx$$

$$= \left[\frac{1}{2} e^{2x} \cdot x \right]_0^2 - \int_0^2 \frac{1}{2} e^{2x} (1) dx$$

$$= \left[\frac{1}{2} e^{4x} x - \frac{1}{2} e^{2x} \right]_0^2 - \frac{1}{4} [e^{2x}]_0^2$$

$$= e^4 - \frac{1}{4} (e^4 - e^0)$$

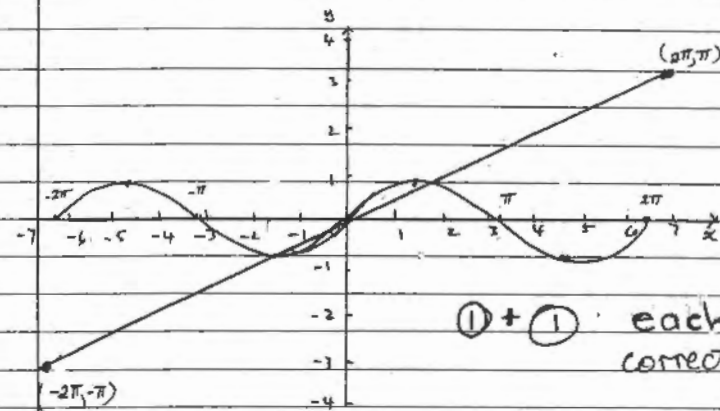
$$= e^4 - \frac{e^4}{4} + \frac{1}{4}$$

$$= \frac{3e^4 + 1}{4}$$

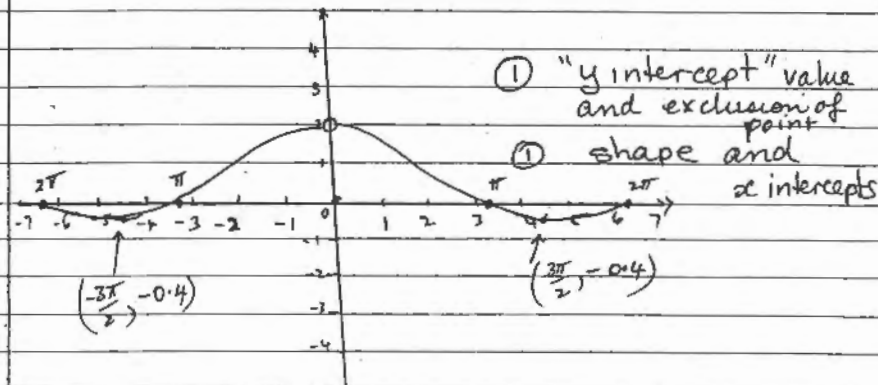
① Integration by parts

① complete integration + substituted correctly

4(b)



① + ① each graph correct



① "y intercept" value and exclusion of point

① shape and x intercepts

4(c) (i) $\frac{x^2}{3} - \frac{y^2}{2} = 1$

$$a^2 = 3$$

$$b^2 = 2$$

$$b^2 = a^2(e^2 - 1)$$

$$2 = 3(e^2 - 1)$$

$$\frac{2}{3} = e^2 - 1$$

$$e^2 = \frac{5}{3}$$

$$e = \sqrt{\frac{5}{3}} \quad e > 0$$

$$\therefore \text{Foci } y=0 \quad x = \pm ae$$

$$= \pm \sqrt{3} \sqrt{\frac{5}{3}}$$

$$= \pm \sqrt{5}$$

$$\text{foci} = (\pm \sqrt{5}, 0)$$

① for 'e' value

① For correct (foci values)

(ii) Equation to tangent at P is:

$$\sqrt{2}x \sec \theta - \sqrt{3}y \tan \theta = \sqrt{6}$$

Equation to perpendicular line is:

$$\sqrt{3}x \tan \theta + \sqrt{2}y \sec \theta = c$$

Normal passes through P

$$\therefore \sqrt{3}(\sqrt{3} \sec \theta) \tan \theta + \sqrt{2}(\sqrt{2} \tan \theta) = c$$

$$\therefore 3 \sec \theta \tan \theta + 2 \tan \theta \sec \theta = c$$

$\therefore$ Equation to normal is:

$$\sqrt{3}x \tan \theta + \sqrt{2}y \sec \theta = 5 \sec \theta \tan \theta$$

① perp equation or gradient

① correct 'c' value and equation of normal

(iii) Equation of tangent

$$\sqrt{2}x \sec \theta - \sqrt{3}y \tan \theta = \sqrt{6}$$

y intercept $x=0$

$$\therefore y = -\frac{\sqrt{6}}{\sqrt{3} \tan \theta} \quad (\tan \theta \neq 0)$$

$$\therefore T = (0, -\frac{\sqrt{2}}{\tan \theta})$$

Equation of normal

$$\sqrt{3}x \tan \theta + \sqrt{2}y \sec \theta = 5 \sec \theta \tan \theta$$

$$x=0 \quad y = \frac{5 \tan \theta}{\sqrt{2}} \quad (\sec \theta \neq 0)$$

$$\therefore N = (0, \frac{5 \tan \theta}{\sqrt{2}})$$

① y coord of T

① y coord of N

$$\begin{aligned} \text{gradient of SN} \\ m_{SN} &= \frac{5 \tan \theta - 0}{\sqrt{2}} \\ &= \frac{0 - \sqrt{5}}{-\frac{\sqrt{5} \tan \theta}{\sqrt{2}}} \end{aligned}$$

$$\begin{aligned} \text{gradient ST} \\ m_{ST} &= \frac{-\sqrt{2}}{\tan \theta} - 0 \\ &= \frac{0 - \sqrt{5}}{\sqrt{5} \tan \theta} \end{aligned}$$

$$= \frac{\sqrt{2}}{\sqrt{5} \tan \theta}$$

$$\therefore m_{SN} \times m_{ST}$$

$$= -\frac{\sqrt{5} \tan \theta}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{5} \tan \theta}$$

$$= -1$$

$\therefore$ SN is perpendicular to ST

$\therefore$ Focus S ($\sqrt{5}, 0$) is on circle with diameter TN (angle in a semicircle is 90°)

Similarly for S' ($-\sqrt{5}, 0$) by symmetry (or further proof)

Alternative Solutions:

(a) Find equation to circle through T and N and test S and S'

(b) Find midpoint of TN and then distances to T, S and S'

① gradients ST and SN

① showing ST perp to SN, geometric reason why S is on the circle.

① Reason for S' on circle (no marks for further algebra)

① Equation of circle
① Test for S on circle
① Test/reason S' on circle

① midpoint of TN and distance to T
① distance to S
① Reason for S' on circle